

MMET 410 – Lab 1

Fall 2026

Tianle Zhang

Please LOG IN: Desktop, Canvas

You can get today's slide from the link:

<https://tianle2000.github.io/teaching/mmet410/>

About class

- Tianle Zhang
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- Recommend to use two computers: One for RSLinx 5000 software. One for reading document.
- Office hour: Wednesday 3:00-5:00 pm 115CA

A detailed floor plan of Thompson Hall, Level One. The plan shows a complex arrangement of rooms, corridors, and stairwells. A red outline highlights a specific area in the lower right quadrant, which includes rooms 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200. A red triangle is placed in room 114. The plan is labeled 'THOMPSON HALL' and 'LEVEL ONE'.

THOMPSON HALL

LEVEL ONE

SCALE: 1"=35'-0"

DATE: 7/21/88

BY: [Signature]

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Grading Policy

- Attend all labs and turn in your lab reports on time
- Lab reports are to be written following the format of sample available on Canvas and must address all question in the lab document
- Lab reports are due by **11:59 PM on the day of your lab section**. However, if you attend the **6:00–8:00 PM section**, your report is due by **11:59 PM the next day**.
- The lab sessions are graded out of 100 points. Here, 70 points are for your practical demo and 30 for your lab reports.

Lab report format

- Title Page – Lab Number, Name and Date
 - The body of the report should include the following for each task:
 - a. Problem Statements – Task Number
 - b. Sequence of operations
 - c. Input and output devices and terminal address assignments
 - d. Ladder logic program (*Not necessary in lab1, but need later*)

Lab Report for Lab 1

- Lab 1: Answer all review questions

(1) What is the workstation chassis size? (How many modules)



(2) Check each box below as you locate the components indicated:

- 1 ☐ Power On/Off switch
- 2 ☐ Power LED
- 3 ☐ Module LEDs



(3) Check each box below as you locate the components indicated:

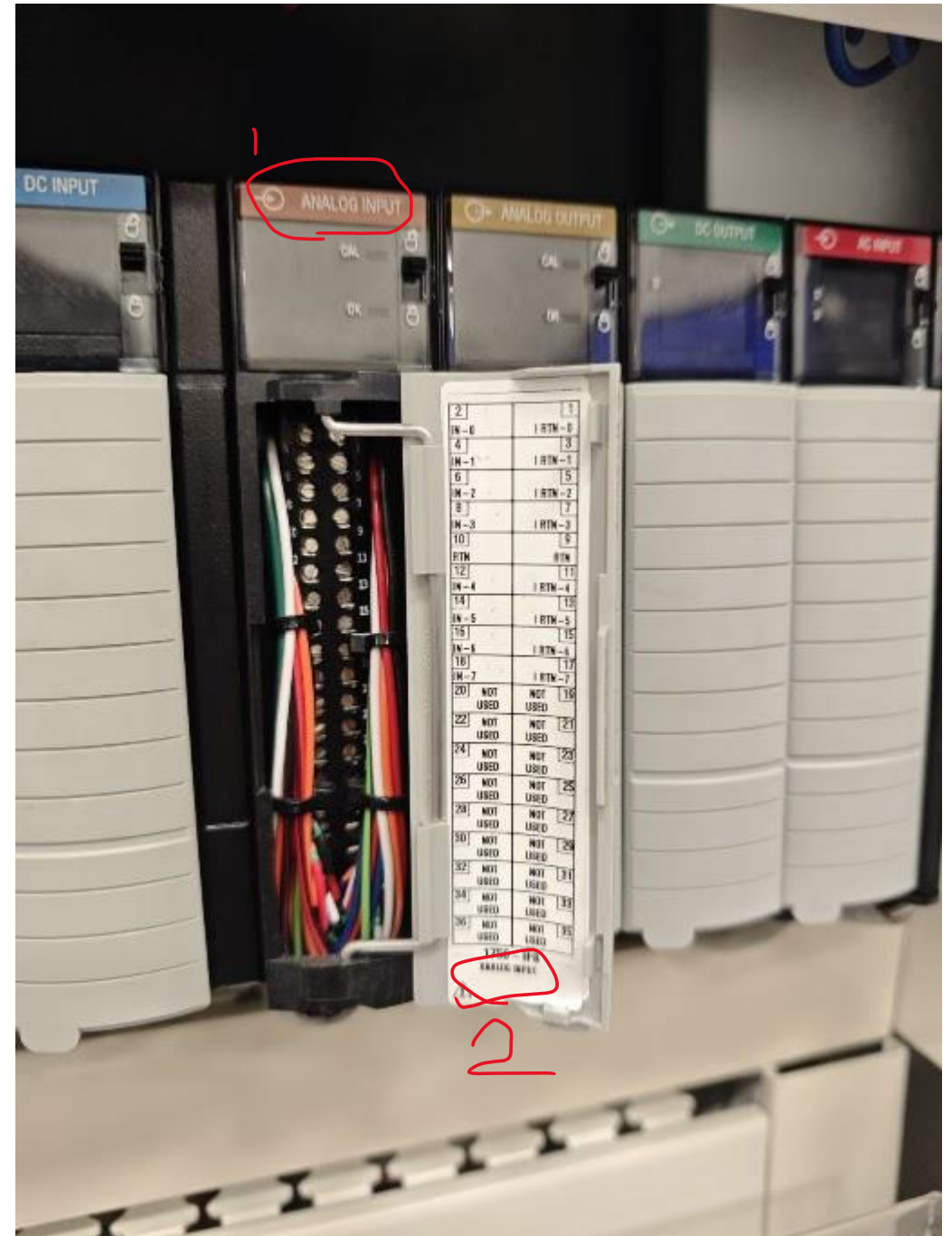
- 1 ☐ Keyswitch and keyswitch positions
- ☐ Battery
- 2 ☐ Serial port

Battery is behind the module with switch



(4) Write the name and catalog number of the workstation hardware in the spaces provided:

- | | |
|------------------|------------------|
| A. Slot 0: _____ | F. Slot 5: _____ |
| B. Slot 1: _____ | G. Slot 6: _____ |
| C. Slot 2: _____ | H. Slot 7: _____ |
| D. Slot 3: _____ | I. Slot 8: _____ |
| E. Slot 4: _____ | J. Slot 9: _____ |



PLC Ladder Logic Input & Instruction

SWITCH ACTION	SWITCH TYPE	INPUT BIT	PLC INSTRUCTION	LOGIC EXAMINATION	RUNG RESULT	OUTPUT STATE
RELEASED	Normally Open (NO)	Bit = 0	XIC 	Examine if Bit = 1 (ON)	✗ False	🔌 De-energize
	Normally Open (NO)	Bit = 0	XIO 	Examine if Bit = 0 (OFF)	✓ True	⚡ Energize
	Normally Closed (NC)	Bit = 1	XIC 	Examine if Bit = 1 (ON)	✓ True	⚡ Energize
	Normally Closed (NC)	Bit = 1	XIO 	Examine if Bit = 0 (OFF)	✗ False	🔌 De-energize
PRESSED	Normally Open (NO)	Bit = 1	XIC 	Examine if Bit = 1 (ON)	✓ True	⚡ Energize
	Normally Open (NO)	Bit = 1	XIO 	Examine if Bit = 0 (OFF)	✗ False	🔌 De-energize
	Normally Closed (NC)	Bit = 0	XIC 	Examine if Bit = 1 (ON)	✗ False	🔌 De-energize
	Normally Closed (NC)	Bit = 0	XIO 	Examine if Bit = 0 (OFF)	✓ True	⚡ Energize

Task 2 to 4

Follow the procedures from **page 3 to page 14** in **410-Lab-2026.pdf**



410-Lab-2026.pdf

Turn on the machine. (Two steps)

1. Red button



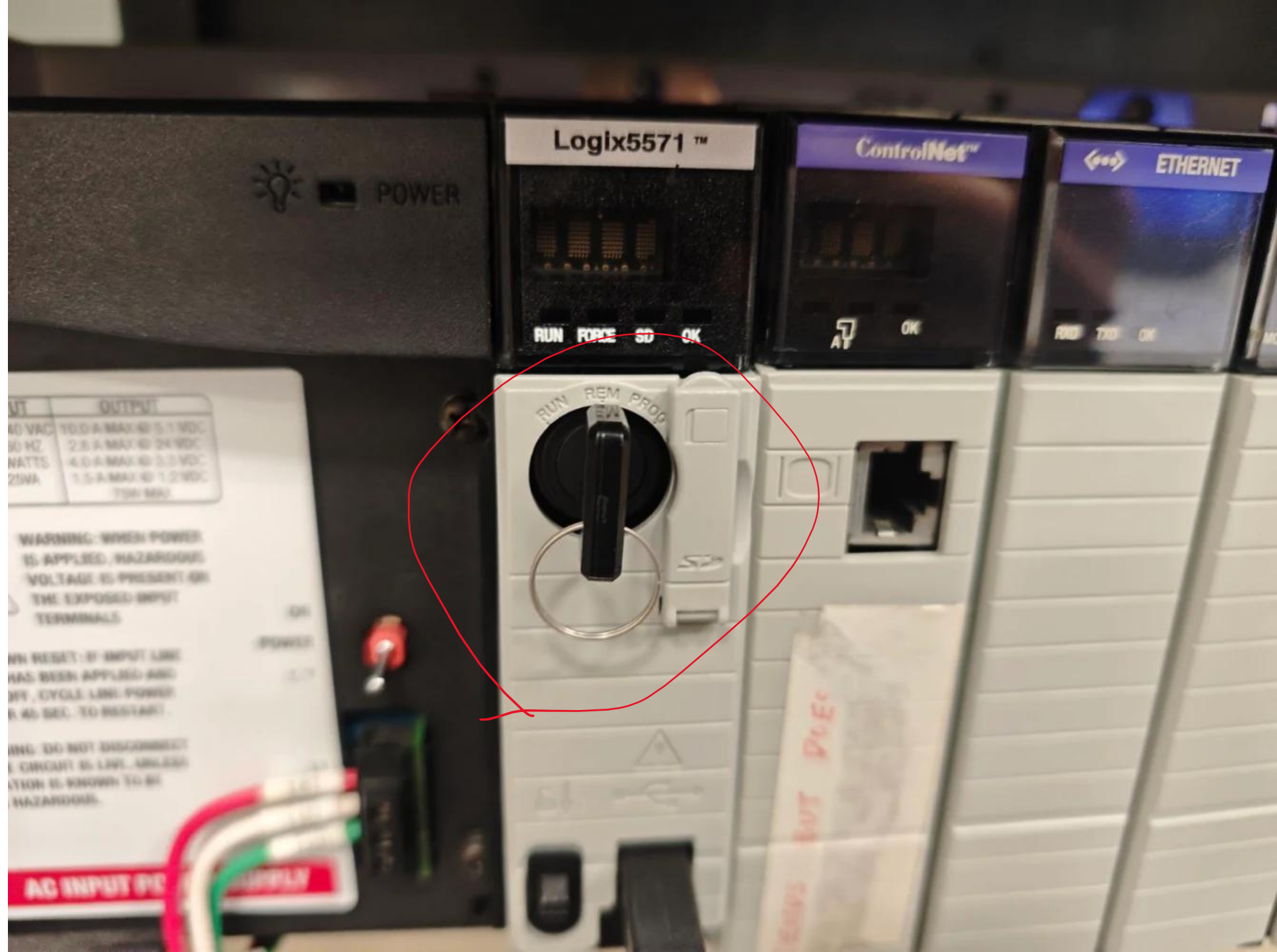
2. Power On/Off switch .

Turn on

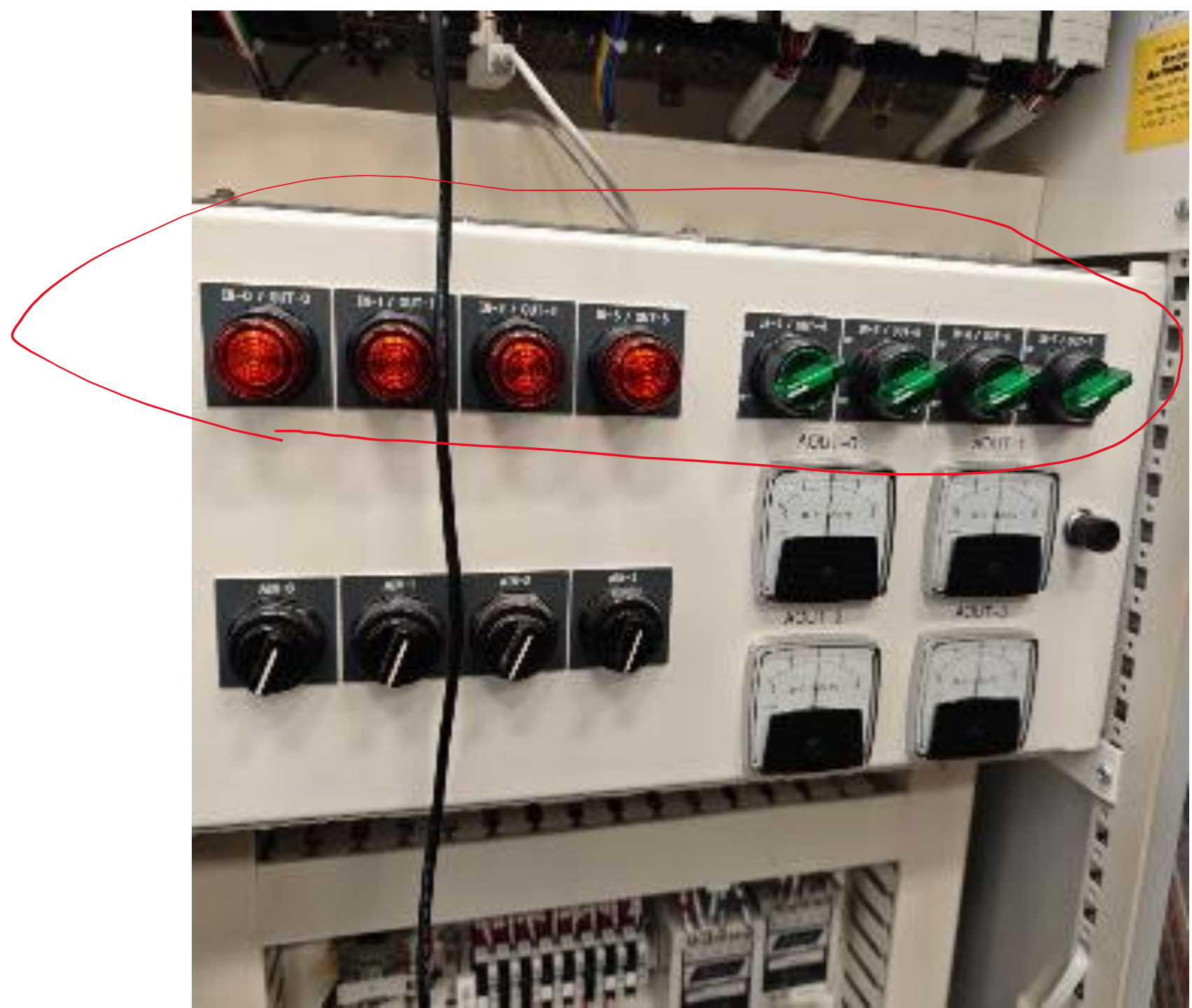


3. Confirm switch key position to REM

Keyswitch and keyswitch positions



Input and output



Connect the software with the hardware

Use the software **RSLogix 5000** (not the emulate on the desktop)

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Create ControlLogix file and finish the I/O Configuration

Suggest: **Save the file under your personal folders**, next time you can use it directly without I/O Configuration again.

Specify what the component in ladder diagram represented in real world

Program Tags

Like a excel sheet

Alias

Local.8.I.Data.5

Local.9.O.Data.1

Write and edit the ladder diagram in software

Flexible way to open the tag library and bounding it with your component.

Lab final:

You will get a random lab you have done this semester

Don't forget to download your project to your own drive.